

Business Development Lab

DP6 - Group 5

-Schedule-

Bianca Ion
Bogdan Pop
Richard Alexander
Jan Kröger

Introduction	2
Chapter 1: User Needs	2
1.1 Rationale for User Needs	2
1.2 Research Design	3
1.2.1 Semi-Structured Interviews	3
1.2.2 Observational Studies	3
1.3 Data Collection	3
1.4 Analysis	4
1.5 Adjusted Customer Needs Hypotheses	5
Chapter 2	6
2.1 Rationale for the prototype	6
2.2 Feature-to-Need-Match (1:1)	7
Table 5 - Feature matching	8
2.3 MVP design	8
2.4 Research design for solution validation	8
2.5 Data collection - solution interviews	9
2.6 Data analysis - solution/need fit	10
2.7 Adjusted solution hypotheses	10
Chapter 3: Financial Model	11
3.2 Cost Model	12
Infrastructure Costs (Year 1)	12
AI API Costs (Year 1)	12
Total Estimated Costs — Year 1	13
3.3 Revenue Model	13
Revenue Projections	14
3.4 Financial Viability	15
Sensitivity Analysis	15
Market Upside & Growth Potential	16
Key Financial Risks & Mitigation Strategies	16
Conclusion	17
Chapter 4	17
4. 1 Unique Value Proposition	17
4. 2 Unfair Advantage	18
4.3 Key Metrics	18
4.4 Channels	19
4.5 How the business side shaped the project	20
4.6 How the technology side shaped the project	20

Introduction

This paper presents the development and validation of SchedulAI, an AI-powered academic planning assistant designed to reduce the cognitive effort students experience when managing their university workload. Rather than focusing solely on productivity or automation, the project is grounded in the insight that students primarily struggle with organizing, prioritizing and coordinating tasks across fragmented systems. Building on user research, prototype design and solution validation, the paper outlines how SchedulAI integrates learning management systems, scheduling tools and AI-driven support into a single platform. Furthermore, it examines how both business and technological considerations shaped the evolution of the product, ensuring alignment between user needs, technical feasibility and financial viability.

Chapter 1: User Needs

This chapter describes our research into student time-management needs: why the problem matters, how we designed the research, what we found, and how our understanding evolved.

1.1 Rationale for User Needs

University students face a uniquely complex planning environment, they are expected to simultaneously manage multiple courses, overlapping deadlines, group projects, extracurricular activities, and personal wellbeing - without the type of planning infrastructure that is in place for professionals. Stress in higher education is well-documented, and poor time management is consistently identified as a leading contributor to academic underperformance and student dropout (Adams & Blair, 2019; Misra & McKean, 2000).

Our motivation arose from direct experience as students at Universiteit Twente. We observed that the core problem is not the volume of work itself, but the planning effort required to organise it. Students frequently struggle to balance academic demands with day-to-day responsibilities (Adams & Blair, 2019): Canvas deadlines are visible but not actionable, and coordinating group work through messaging applications such as WhatsApp, proves to be fragmented and time-consuming. The resulting cognitive overload leads to last-minute work, missed deadlines, disrupted sleep, and a persistent feeling of guilt even during legitimate rest time. We identified an opportunity for an AI-powered planning assistant that reduces mental effort while keeping students in control of their schedule.

The Dutch student market alone comprises over 300,000 individuals (Statista, 2022). Canvas is used by more than 30 million users across 100+ countries (Instructure, 2024), demonstrating the scale of the addressable market for a solution that integrates with existing learning management systems.

1.2 Research Design

To validate our assumptions and identify genuine user needs, we employed two complementary research methods, being a series of semi-structured interviews and structured observational studies.

1.2.1 Semi-Structured Interviews

We conducted seven interviews with students within the Universiteit Twente library and chose to use a convenience sampling method. The only selection criteria were that participants had to be current students with a baseline level of technical knowledge. Interviews were conducted online via Microsoft Teams or by phone call and lasted approximately 20–30 minutes each. The interview guide consisted of 11 core questions across five key themes that we had identified to be pertinent to the research objectives, and with space for unstructured follow-up questions:

- Academic workload and stress levels
- Time management practices and deadline tracking
- Extracurricular activities and personal wellbeing
- Learning platform experience (Canvas/Moodle)
- Subscription behaviour and willingness to pay for an AI tool

1.2.2 Observational Studies

Three structured observational studies were conducted with participants who were current university students, enrolled in at least two courses, and involved in at least one group project. The observations took place in a private library room setting, the participants were asked to bring their own laptops, this was to ensure ecological validity and they were also asked to “think aloud” while performing the realistic tasks given. No assistance was provided during the study. The three scenarios were:

- Observational Study 1: Log into LMS and identify all deadlines for one course, then add tasks to a personal calendar.
- Observational Study 2: Locate all relevant course materials for an upcoming exam using only the LMS.
- Observational Study 3: Three participants jointly organise a group project meeting, agreeing on a time and location using their usual tools.

1.3 Data Collection

Seven student interviews were conducted covering a range of study levels (bachelor and master) and lifestyles. The following table summarises key findings from each interview:

Interview	Key Profile	Main Pain Point	Attitude to AI Planning
-----------	-------------	-----------------	-------------------------

1	Irregular workload, misses deadlines, plays tennis 3×/week	Cannot estimate task duration; Canvas navigation on Moodle is confusing	Open if UI is comprehensive; useful for scheduling and sleep
2	Disciplined, prefers to start early; group project friction	Teammates work last-minute; Canvas inconsistent across professors	Helpful for group work coordination; comparative price mindset
3	Procrastinates; going to gym 4×/week; stressed by workload	Unsure how long tasks take; information on Canvas hard to find	Enthusiastic — "exactly the type of thing I need"; pays €5–20/month
4	Disciplined; active socially; irregular sleep due to deadlines	Lack of preparation causes stress; over-reliance on group motivation	Supportive — sees AI as assistant for day-to-day scheduling
5	Manageable schedule; deadlines during exam weeks overwhelming	Pre-created schedules break down during busy periods	Sees value in a holistic view of university workload
6	Very busy; procrastinates; stressed; parties and sport	Canvas feels too casual; important info hidden in documents	Finds it cool; would pay up to €25/month; values quick document search
7	Heavily involved in music (18–22 hrs/week); well-organised	Information scattered across LMS; needs effort to get overview	Curious; concerned about differentiation from existing tools

Table 1 - Interviewee profiles

Three observational studies revealed the following friction points:

- Study 1 (Deadline tracking): LMS deadlines are visible but not actionable. Participants manually created vague calendar entries with no time estimates, creating high cognitive load.
- Study 2 (Material organisation): Course content scattered across LMS sections. Students navigated multiple tabs, expressed uncertainty about which materials mattered, and manually took notes to summarise key points.
- Study 3 (Group coordination): No shared overview of availability. Scheduling a single meeting required repeated WhatsApp messages, uncertainty about room availability, and significant back-and-forth.

1.4 Analysis

Across all research activities, five core problem clusters emerged:

Problem Cluster	Evidence	Severity
Poor time estimation	Students cannot estimate the amount of time needed for assignments;	High

	underestimation causes last-minute stress	
Canvas/LMS disorganisation	Information is scattered; deadlines visible but not actionable; inconsistency across professors	High
Group scheduling inefficiency	Too many messages to schedule a meeting; no shared calendar overview; back-and-forth coordination	High
Sleep inconsistency	Irregular sleep schedules; stress spikes near deadlines disrupt sleep routines	Medium
Guilt during free time	Students feel guilty when not working, even during legitimate rest, due to unclear task load	Medium

Table 2 - Interview core problems

A competitor analysis confirmed these needs are underserved. The closest existing solution, Shovel, integrates with Canvas and Moodle but only imports exam dates, it lacks both study support and group scheduling. MyStudyLife allows class and assignment tracking but requires manual data entry and has no LMS integration. General AI planners (e.g. Motion, Reclaim.ai) offer strong automation but are not designed for students: they lack LMS integration, exam-aware scheduling, and support for academic grading periods. ScheduAI addresses all these gaps in a single, student-centred application.

Key insight: The primary problem students face is not the volume of work - it is the mental effort and stress required to organise that work. Students need cognitive relief, not just another tool to manage (Adams & Blair, 2019).

1.5 Adjusted Customer Needs Hypotheses

At the outset we defined three customer needs hypotheses. After our research, we adjusted them as follows:

Original Hypothesis	Research Finding	Adjusted Hypothesis
50% of users suffer sleep and wellbeing issues from university-related stress	Confirmed across all 7 interviews. Irregular sleep patterns and guilt during free time were consistently reported.	Validated and strengthened. Target users experience cognitive overload and guilt even during rest, not just during deadline periods. The core need is stress reduction through planning clarity, not just time management.
65% of users are frustrated with their current LMS and its	Confirmed. All interview and observation participants reported friction with Canvas:	Validated. The frustration is structural: students must perform "mental translations"

effectiveness for learning habits	scattered materials, inconsistent course structure, and vague deadline presentation.	between LMS information and personal calendar planning. An automated LMS-to-calendar bridge is the primary need, not a better LMS.
70% of potential users are willing to pay a subscription fee for an AI tool that helps organise university work and extracurricular activities	Partially validated. Willingness to pay exists but is lower than expected. Most students prefer a freemium model and will pay only if clear value is demonstrated. Price sensitivity is high among students.	Revised. The business model must lead with a compelling free tier. Revenue depends on demonstrating unambiguous value before asking for payment. A freemium model with premium AI features is more appropriate than a fully paid product.

Table 3 - Customer needs

Chapter 2

In this chapter, we describe the design and rationale for the SchedulAI prototype, the research design for validating the solution, the data we collected from the interviews, and our updated hypotheses.

2.1 Rationale for the prototype

After interviewing and observing students, we managed to get some insights into their life and habits, which helped us in validating problems, and to better understand the pain points and their needs. From the interviews, the core problem clusters identified are:

Problem cluster	Evidence from research	Severity
Poor time estimation	Students cannot estimate the amount of time needed for an assignment	High
Canvas organization	Information is scattered and the deadlines can be unclear	High
Group scheduling inefficiency	Too many messages to schedule a meeting, no shared overview	High
Sleep inconsistency	Irregular sleep, stress spikes near deadlines	Medium
Guilt during free time	Students feel guilty when not working	Medium

Table 4 - Problem severity

This analysis shows that most students suffer from a mixture of both uncertainty and cognitive overload, however neither necessarily stem from having too much work. The

fragmented tools and the lack of structured planning make their life stressful. For the majority, the stress is a planning problem that our product helps in correcting.

Based on problem clusters we identified, we designed SchedulAI, an integrated academic planning assistant that connects university platforms with personal calendars, aiming to ease student life, manage workload efficiency and assist in understanding the school materials.

[Demo](#)

The goal of the MVP is to present to the potential users a tangible solution that would allow us to see how it addresses their needs.

The prototype is created and hosted on github pages and is a mock-up that simulates the user experience they would have in the real application. At this stage, this was a better solution for us to validate whether the technology works and is appealing and useful to students, with a low development cost and zero risk of technical failures that could distract us from the research we were doing.

2.2 Feature-to-Need-Match (1:1)

Every feature in the prototype is derived from a customer problem we identified. The following table shows the mapping:

Customer problem	SchedulAI feature	How it solves the issue
Students do not know how long a task takes	AI time estimation engine	AI predicts realistic work duration based on assignment difficulty, size and past data
Canvas deadlines are visible but not actionable	Automatic synchronization	Imports deadlines & creates time-blocked study sessions
Information is scattered	Smart content hub	Aggregates slides, PDFs, deadlines into one dashboard
Group scheduling is inefficient	Group synchronization	Optimal meeting times suggestions based on all calendars
Students procrastinate	Smart priority suggestions	The AI suggests what to start today based on urgency & difficulty
Students feel guilty when resting	Protected free time mode	Allocates rest time deliberately
Irregular sleep schedule	Sleep aware planning	Avoids late-night study blocks

Table 5 - Feature matching

2.3 MVP design

The SchedulAI prototype consists of five core screens accessible via a bottom navigation bar: Home/Dashboard, Calendar, AI Chat, Subjects, and Groups. The onboarding flow takes the user through three set-up steps before reaching the main application:

- **Course Selection:** The user selects which Canvas courses to include in AI scheduling. This gives the user immediate control and prevents unwanted courses from cluttering the schedule.
- **Study Preferences:** The user selects their preferred study time (Morning / Afternoon / Evening) and whether to include personal activities. This enables personalised scheduling.
- **AI Scheduling Mode:** The user chooses between "Assisted" mode (AI suggests, user approves) or "Automatic" mode (AI schedules autonomously). This addresses the trust gap identified in interviews — many users want oversight before committing to AI-generated plans.
- **Dashboard:** Shows an AI suggestion banner with the most urgent task, today's schedule, and upcoming deadlines. Acts as the daily command centre.
- **Calendar:** Full week/month view distinguishing AI-generated events from user-created ones.
- **Subjects tab:** Centralised hub for each course's materials, assignments, and a subject-specific AI chat for course content questions.
- **Groups tab:** Displays group project members, shows AI-analysed availability, and allows members to vote on proposed meeting times.
- **AI Chat:** Global assistant where users can ask scheduling, content, or planning questions in natural language.

2.4 Research design for solution validation

We conducted 10 interviews, with which we want to validate/invalidate our ideas and to make sure our potential clients, the students, appreciate and find useful the application and all the features we implemented with it. Each interview followed a five-phase guide:

- **Phase 1 - Context** - focuses on exploring the user's habits and current behaviour;
- **Phase 2 - Problem confirmation** - aims to make the user realise the problems, frustrations and emotions he has concerning his habits and behaviour;

- Phase 3 - Prototype presentation - introduces the concept and gathers initial reactions to the features we implemented;
- Phase 4 - Pricing and value - evaluates the perceived benefit of the solution by asking about willingness to pay, preferred pricing models and expectations for free versus paid features;
- Phase 5 - Closing - tests real adoption intent by identifying barriers, missing elements and whether users would genuinely try to replace their current tools with this solution.

2.5 Data collection - solution interviews

The following table summarises the findings we had from the solution interview:

Profile	Standout Feature	Concern	WTP
Master student, works part-time, disciplined	AI summarisation	Does not need automation; redundant vs existing tools	None — too organised to need it
Master student, works part-time, uses MS Teams	AI summarisation; hardware integration idea	Needs seamless integration; AI can't estimate time accurately	None — expects it in existing tools
Bachelor, procrastinates during exam weeks	Auto-scheduling; AI chat	Wants control over generated schedule	€10/month; prefers free tier first
Bachelor, structured monthly planner, busy schedule	Deadline auto-import; task integration	Wants to estimate task durations himself	€10/month; wants free tier first
Bachelor, intuitive planner, procrastinates	Auto-scheduling; personal time feature	Would not use for big tasks like full reports	€10/month; expects AI chat as paid feature
Overwhelmed, fragmented workflow across multiple tools	Automatic scheduling (eliminates planning effort)	Needs flexibility; not rigid structure	€5–10/month; expects free variant
Chaotic, minimal tools, low motivation for new apps	No clear standout — sceptical	Automation reduces task awareness	Unwilling to pay; free only
Disciplined but crams; pressure-driven	Automatic scheduling for stress reduction	AI cannot understand individual skill level	€5–15/month for adaptive features
Moderately organised; misses deadlines due to distraction	Reminders and scheduling support	Wants support tool, not autonomous system	Unwilling to pay; low monetisation segment

Low-structure but open-minded; forgets tasks	Auto-scheduling; group coordination	Flexibility concerns in social contexts	~€10/month; needs solid free version
--	-------------------------------------	---	--------------------------------------

Table 6 - Interview findings

2.6 Data analysis - solution/need fit

Summarising our 10 interviews, we found these common trends among our participants:

1. There are users who reject AI for estimating their time. We can improve this by learning from the user behaviour over time, letting users adjust estimates, and showing confidence levels.
2. Automatic scheduling, platform integration and group meeting suggestions are very valued features. Most users give extra value to AI summarisation, meaning we could make it a core feature and improve it with adjustable summary length, exam mode summarisation, or personalised summaries.
3. The product should not control students' schedules, but rather reduce mental effort while keeping them in control. It should also be simple to set up and use.
4. A part of the students want to track the time spent on entertainment, since they do not want to exclude it from their schedule, but are scared of losing track of time.
5. Pricing expectation is low, with the majority of students choosing a freemium model and will pay only if clear value is created.
6. The key insight is that the main problem is not organising, but the effort and stress required to do that.

2.7 Adjusted solution hypotheses

The following table evaluates each hypothesis against the interview findings:

Hypothesis	Target	Result	Verdict
8/10 students agree that AI time estimation reduces stress	≥80%	Not supported. Most users distrust AI time estimation, preferring personal control or wanting AI to learn from their behaviour over time.	Revised: Reposition feature as "adaptive estimate with user override" rather than autonomous prediction
7/10 students prefer automatic Canvas synchronisation over manual planning	≥70%	Strongly supported. Canvas sync was the most consistently valued feature across all profiles.	Confirmed: Core free feature — highest priority for full implementation

6/10 students would use the group scheduling tool regularly	≥60%	Supported. Group coordination is a genuine pain point. Users see immediate value in AI-analysed meeting slots.	Confirmed: Include in MVP; consider this a key differentiator vs competitors
7/10 students see value in paying €5–15/month	≥70%	Partially supported. Willingness to pay exists in the €5–10 range, but most require a strong free tier first. Pure subscription model is unlikely to convert.	Revised: Freemium model with free scheduling and paid AI features
8/10 students say the app reduces cognitive overload	≥80%	Supported. Students consistently identified reduced planning effort as the primary benefit, aligning with our core value proposition.	Confirmed: "Cognitive relief" should be the central marketing message

Table 7 - Solution hypothesis

Chapter 3: Financial Model

This chapter presents a cost model, revenue model, and financial viability analysis for SchedulAI. All figures are based on documented assumptions derived from market research, interview pricing data, and standard SaaS cost benchmarks for early-stage startups. As SchedulAI is currently at prototype stage, these projections represent a realistic first-year scenario.

3.1 Financial Summary Dashboard

Quick overview of SchedulAI's Year 1 financial position:

Metric	Year 1 Value
Total Operating Costs	€5,505
Projected Annual Revenue	€5,535
Break-Even Month	Month 6-7
Break-Even Users Required	443 total (62 paying)
Year-End User Base	900 total (126 paying)
Net Position (Year 1)	+€30 (near break-even)

Table 8 - Financial Dashboard

3.2 Cost Model

SchedulAI's cost structure in the early stage consists of three categories: infrastructure, AI API costs, and operational expenses. Development costs are treated as opportunity cost rather than cash outlay, as the founding team of four students builds the product themselves.

Infrastructure Costs (Year 1)

Cost Item	Description	Monthly (€)	Annual (€)
Cloud hosting (AWS/Firebase)	Backend servers, database, storage, auth	120	1,440
Domain & SSL certificate	schedulai.io domain, SSL	5	60
Development tools	Figma, GitHub Pro, CI/CD pipeline	25	300
Customer support tools	Basic help desk / chat widget	50	600
Legal / compliance	Privacy policy, GDPR compliance consultation	15	180
Sub-total — Infrastructure		215	2,580

Table 9 - Infrastructure costs

Assumption: Cloud costs scale with user base. At 1,000 monthly active users, AWS/Firebase costs are approximately €80–150/month based on standard pricing for read-heavy apps with moderate compute needs.

AI API Costs (Year 1)

SchedulAI's AI features (document summarization, AI chat, scheduling suggestions) depend on a large language model API (e.g., OpenAI GPT-4o-mini or equivalent). Costs are calculated based on estimated usage per user per month.

Scenario	Monthly Active Users	Avg. Queries/ User/Month	Est. Cost/User (€)	Monthly AI Cost (€)
Conservative (free tier only)	500	30	0.10	50
Base case (500 free + 100 paid)	600	50	0.15	90

Growth (1,000 free + 200 paid)	1,200	60	0.18	216
--------------------------------	-------	----	------	-----

Table 10 - API costs

Cost calculation: Using the pricing from GPT-4o-mini(€0.15 per 1M input tokens, €0.60 per 1M output tokens), with 1,500 input tokens and 500 output tokens for every query, the average cost per query is €0.0005. For the free tier with 30 queries/month the cost will be €0.015/month, while paid users have 80 queries/month the cost will be €0.04/month each.

Total Estimated Costs — Year 1

Category	Annual Cost (€)
Cloud infrastructure	2,580
AI API costs (base case)	1,080
Customer support tools	600
Domain, tools, & legal	540
Marketing (campus flyers, social ads)	1,500
Payment processing fees (est. 3% of revenue)	165
Contingency buffer (10%)	545
TOTAL	7,010

Table 11 - Total estimated costs

This represents a lean cost structure achievable without external funding in Year 1, as development labor is contributed by the founding team. Monthly fixed costs average €460 (€5,505 ÷ 12), plus variable costs that scale with usage.

3.3 Revenue Model

SchedulAI adopts a freemium subscription model, consistent with user preferences expressed across both DP2 and DP4 interviews. The model consists of three tiers:

Tier	Price	Features Included	Target Segment
------	-------	-------------------	----------------

Free	€0/month	Canvas sync (2 courses), basic AI scheduling, deadline reminders	All students — primary acquisition channel
Student Premium	€5.99/month or €49/year	Unlimited Canvas sync, AI summarization, group scheduling, AI chat, protected free time mode	Students who experience clear value from free tier
Team/Institutional	€8.99/user/month (min. 5 users)	All premium features + group admin dashboard, priority support	Student associations, study groups, institutional licensing

Table 12 - Revenue model

Annual pricing note: The €49/year option (€4.08/month effective rate) is projected to attract 30% of premium subscribers, based on typical SaaS annual vs. monthly subscription splits. This is factored into the revenue projections below.

Revenue Projections

Revenue projections were created for a 12-month window, starting with the University of Twente student base of approximately 10,000 students and expanding through campus marketing and referrals from students. Also, an annual churn rate of 10% is applied..

Period	Free Users	After Churn	Premium Users	After Churn	Monthly Revenue (€)	Cumulative (€)
Month 1-3	150	146	15	15	82	246
Month 4-6	400	390	40	39	213	885
Month 7-9	700	682	80	78	426	2,163
Month 10-12	1,000	974	140	136	747	4,404
Year-End	1,000	974	140	136	747/mo	€4,404

Table 13 - Revenue Projection

Institutional Revenue: Conservative projection includes 1 student association pilot (15 users at €8.99/month) starting Month 10, generating an additional €405 in Year 1 revenue. Total Year 1 revenue including institutional: €4,809.

Conversion rate assumption: 14% free-to-paid conversion, consistent with the interview finding that a meaningful minority expressed clear willingness to pay, influenced by high price sensitivity.

Churn methodology: 10% annual churn rate is equivalent to 0.87% monthly churn. This accounts for students graduating, changing study patterns, or switching tools. The churn rate is conservative compared to typical student SaaS products (15-20% annual churn).

3.4 Financial Viability

SchedulAI reaches break-even when monthly revenue covers monthly costs.

Metric	Value
Monthly fixed + variable costs (base case)	~€460/month
Average revenue per paying user (blended monthly/annual)	€5.49/month
Payment processing cost per user (~3%)	-€0.16/month
Net revenue per paying user	€5.33/month
Break-even paying users required	87 users
Break-even total users required (at 14% conversion)	~620 users
Estimated time to break-even	Month 7-8

Table 14 - Break-even estimation

In order to get to break-even the application will need around 620 active users. The University of Twente has roughly 10,000 students. Getting the 620 active users needed after 7-8 months is a low target.

Sensitivity Analysis

The table shows how different conversion rates modify the time to break-even:

Scenario	Conversion Rate	Months to Break-Even
----------	-----------------	----------------------

Pessimistic	10%	Month 9-10
Base Case	14%	Month 7-8
Optimistic	20%	Month 5-6

Table 15 - Sensitivity analysis

Even in the pessimistic scenario, SchedulAI reaches break-even within the first year, demonstrating the robustness of the business model.

Market Upside & Growth Potential

The Netherlands has more than 300,000 university students. Canvas is used by more than 30 million users globally. Even at 0.1% of Canvas users, SchedulAI would serve 30,000 users — generating approximately €252,000/year in premium revenue at a 14% conversion rate (4,200 paying users × €5.99/month × 12), substantially exceeding operating costs.

Year 2 projection: Based on Year 1 momentum and expanding to 2-3 additional Dutch universities, Year 2 projections target 5,000 total users (700 paying) generating €50,000+ annual revenue, while operating costs remain under €15,000 with scaled infrastructure and increased but still-lean AI costs.

Key Financial Risks & Mitigation Strategies

Risk	Impact	Mitigation Strategy
AI API cost growth as user base scales	Could compress margins at high scale	Usage caps for free tier, optimize prompts, negotiate volume discounts at scale
Growth exceeds revenue runway	Temporary negative cash flow	Incremental funding from university startup grants (€5,000-10,000 available), control growth pace
Lower-than-expected conversion rate	Delayed break-even (10% → Month 9-10)	A/B test pricing, add compelling premium features based on user feedback, targeted conversion campaigns

Price sensitivity among students	Resistance to paid tier adoption	Strong free tier creates trust, student discount campaigns, emphasize time-saved ROI
----------------------------------	----------------------------------	--

Table 16 - Financial Risks & mitigation strategies

These risks are substantially mitigated by the lean cost structure, bootstrapped development approach, and freemium model that reduces barriers to initial adoption while creating clear upgrade paths for engaged users.

Conclusion

SchedulAI's financial model demonstrates strong viability with conservative assumptions. The combination of lean infrastructure costs, favorable AI pricing, and a proven freemium SaaS model creates a path to break-even within 7-8 months and profitability by year-end. With a large addressable market and multiple expansion opportunities (institutional licensing, geographic expansion, feature upsells), SchedulAI is positioned for sustainable growth beyond the initial launch phase.

Chapter 4

In this chapter, we will talk about all the unique value propositions as well as the unfair advantage of SchedulAI. Furthermore, we will discuss the main channels we could use to market our product and the key metrics to measure success. Finally, we will also discuss how the business and technology side of the project have shaped the project along the way.

4. 1 Unique Value Proposition

SchedulAI's unique value proposition is not just automating but reducing the workload required for planning tasks and academic tasks while keeping the students in control. The interviews made clear that students often do not struggle with the university content itself but with the mental side of organising, prioritising, prioritizing and coordinating different tasks and juggling their hobbies and free time at the same time. Traditional tools like digital calendars and Canvas require manual translation of deadlines into schedules. SchedulAI addresses this by automatically integrating students' university data from Canvas into a structured schedule. Furthermore, SchedulAI provides AI generated summaries and an AI chat about the university subjects. SchedulAI also helps with scheduling group meetings by checking availability across the group members and creating suggestions.

These features allow for adaptive planning. The AI makes suggestions, but the user is ultimately the one to add them to their schedules. Users stay in control and get decision support. This is supported by our interview findings, as a key insight was that users do not want full AI control.

4. 2 Unfair Advantage

ScheduleAI's unfair advantage lies in the combination of integration, personalization and strong focus on reducing student's cognitive load rather than simply improving productivity. Many existing tools offer parts of the solution such as scheduling or AI support but none combine these elements in a way that directly addresses our identified core problem. A key advantage is the direct integration with Canvas. Students do not need to manually put in their deadlines, course information, etc. into ScheduleAI. The Canvas integration automatically synchronizes everything important and uses that information in the schedules. This removes a major source of mental effort for students. In addition, ScheduleAI creates an ecosystem by combining multiple features that are usually spread across different tools. Scheduling, AI summaries, group coordination and course material access are all in a single platform. This reduces the need to switch between applications and increases the overall value for the users. Finally, the product is built on validated user insights. Our design approach makes use of these user insights and makes the concept more difficult for competitors to replicate without similar research and understanding of the user needs.

4.3 Key Metrics

User activation	· % of users connecting the Canvas Account · % of users generating their first AI schedule · % of users using the AI Chatbot
Engagement metrics	· Daily/Weekly active users · Number of schedules generated per user · Number of AI chats by user
Behavioural impact	· Reduction of missed deadlines · User-reported stress level · User-reported planning effectiveness · % of AI generated planning suggestions accepted
Retention	· % of users continuing after generating their first AI schedule · Week 1 - Week 5 retention
Monetization metrics	· % of users converting to paid subscription

	· Most used features per user
--	-------------------------------

Table 17 – Key Metrics

ScheduleAI's key metrics focus on measuring whether the product actually reduces cognitive load and improves how students manage their time. The main goal of ScheduleAI is not just having as many users as possible but serving an actual purpose for our users. We want to test it with activation, engagement, behavioural, retention and monetization metrics. This spread of metrics will ensure a big picture view on our results.

One of the most important metrics is user activation. This includes how many users connect their Canvas account and generate their first AI-generated schedule. This indicates whether users understand the value of the product and are willing to include it in their workflow. Furthermore, we want to see if the AI Chat is a feature that the users will use after activating their product.

Engagement metrics include how frequently users interact with the application. This will check the number of daily and weekly users. Also, the engagement metrics will quantify the number of schedules generated per user as well as the number of AI chats done by the user.

Another key area is behavioural impact. Since the goal is to reduce stress and planning effort, it is important to measure outcomes such as a reduction in missed deadlines, user reported decreases in stress and user reported planning effectiveness. Lastly, the behavioural impact can be tested by measuring the amount of AI suggested schedules the user approves. This is also interesting to compare over time. These metrics directly reflect whether the product is solving the problems identified in the interviews.

Retention is also important as students are unlikely to continue using a tool that does not provide any value for them. Tracking the number of users that stay with the product after their first AI generated schedule tests the initial value of the product. Testing the retention rate after a few weeks of usage is also going to reveal trends in prolonged usage patterns.

Finally, monetization metrics are important as our main target group are students. Students tend to only spend money on things that add real value to their lives. Here we can look at the conversion rate from free to paid users and which of the specific features get users to upgrade. These metrics help ensure that the business model aligns with user expectations and perceived value.

4.4 Channels

ScheduleAI's channels focus on reaching students in a way that fits naturally into their existing workflow and that is not difficult to adopt. Students are unlikely to actively search for new productivity tools. The channels we want to utilise are designed to integrate into platforms and environments that students are already a part of and use. A primary channel is the direct integration with learning management systems such as Canvas. By connecting

directly to the platform that students already use for their university work, SchedulAI can serve as an extension of their current workflow and not as an additional tool. Students will automatically start using our SchedulAI without greatly changing any habits.

Mobile and web applications also play an important role as distribution channels. Students mostly use their phones to manage their schedules which suggests they prioritize a mobile version. A web version could add to this with more detailed interaction.

Another important channel is word of mouth within student communities. Students often rely on recommendations from peers. This is especially true when it is about university work or other related topics. Study groups, project teams and student associations can therefore play a big role in gaining popularity.

The freemium model itself also acts as a channel. By offering a free version with core features, students can try the product without any commitment. This is very important as we observed a low willingness to pay in the interviews. This allows the product to show its value before asking users to upgrade.

Finally, social and digital platforms like Instagram or TikTok can be used to reach students quickly and effectively. Content focused on productivity, student life and stress reduction aligns well with the problems SchedulAI addresses and can attract potential users. The user base for social media widely overlaps with our intended user base which makes this a very important channel.

4.5 How the business side shaped the project

From the business side, the interview insights directly shaped the technical priorities. One important example is the reduced focus on AI time estimation. It was originally considered as a core feature but the interviews revealed that many students do not trust AI to accurately estimate how long their assignments take. This led us to pay more attention to features that were revealed to be important to the users like automatic scheduling, AI summaries and a single platform integration. Another key influence was the importance of flexibility and user control. The interviews revealed that users do not want the AI to take full control over their scheduling. This made us change the AI's role to be more supportive and suggest rather than having full control. The AI suggests schedules and actions but the user makes the final decisions. These changes made it necessary to have a system that allows for easy adjustments and approvals rather than full automation. Additionally, the business model influenced the technical structure of the product. Using a freemium model requires a clear separation between free and premium features. This led to a design with the core features available in the free-tier but AI driven high computation features behind a paywall.

4.6 How the technology side shaped the project

The technology side also shaped this project. The use of AI models and cloud infrastructure has many unexpected costs which makes a free product unsustainable. A paid Premium version is necessary to sustain the product. Furthermore, the complexity of integration with platforms like Canvas highlight the relevance of partnerships with universities and potentially

other LMS providers. Finally, technical possibilities also influenced the product's features. Some ideas like a highly accurate time prediction are difficult to implement reliably and require large amounts of data. This led to a shift in the business focus towards more user-focused input and other AI integration features.

References

Adams, R. V., & Blair, E. (2019). Impact of time management behaviors on undergraduate engineering students' performance. *SAGE Open*, 9(1). <https://doi.org/10.1177/2158244018824506>

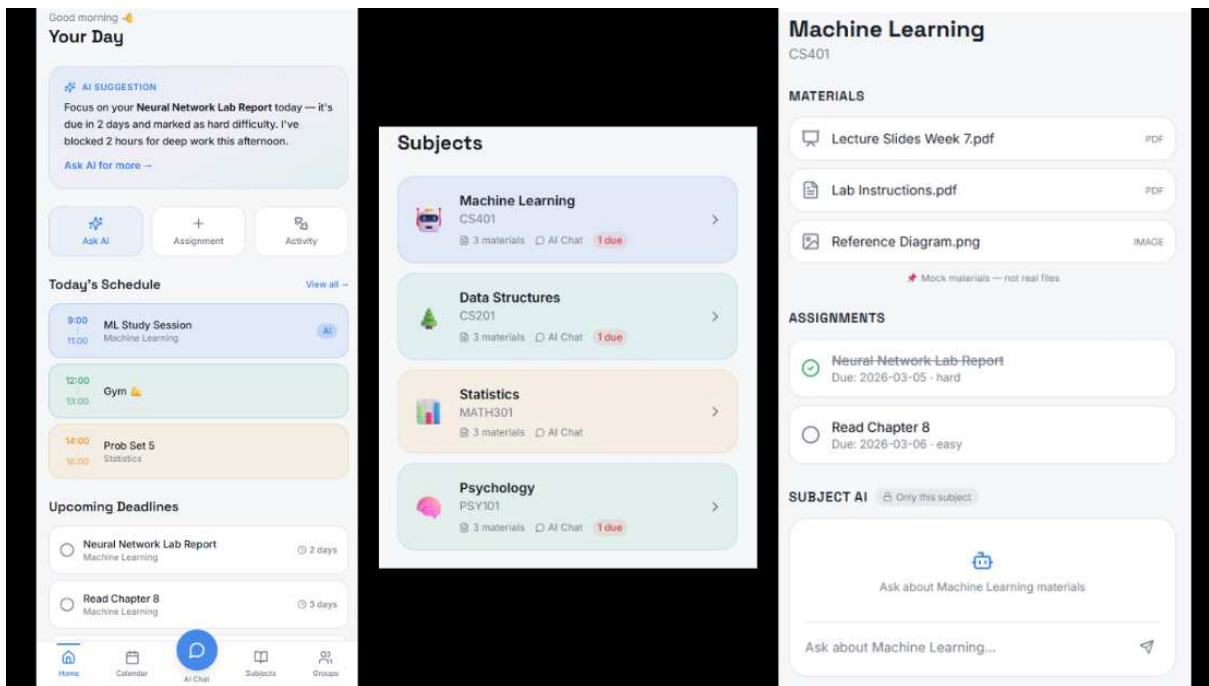
Instructure. (2024). Canvas by Instructure: World leading LMS for teaching & learning. Retrieved from <https://www.instructure.com/canvas>

Misra, R., & McKean, M. (2000). College students' academic stress and its relation to their anxiety, time management, and leisure satisfaction. *American Journal of Health Studies*, 16(1), 41–51.

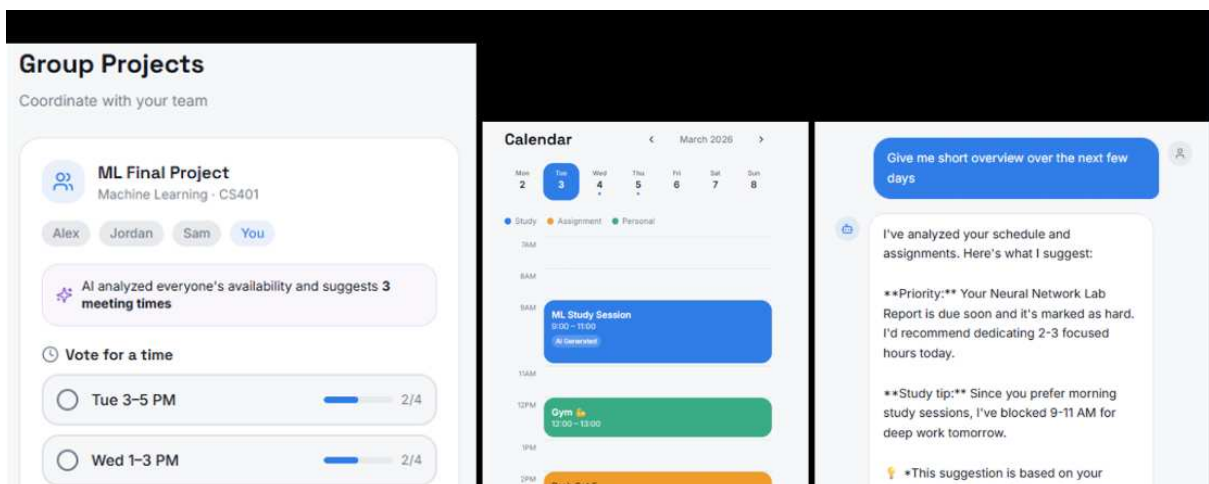
Statista. (2022). Number of university students in the Netherlands 2016–2022. Retrieved from <https://www.statista.com/statistics/782018/number-of-university-students-in-the-netherlands/>

Appendicies

Appendix A:



Appendix B:



Appendix C:

